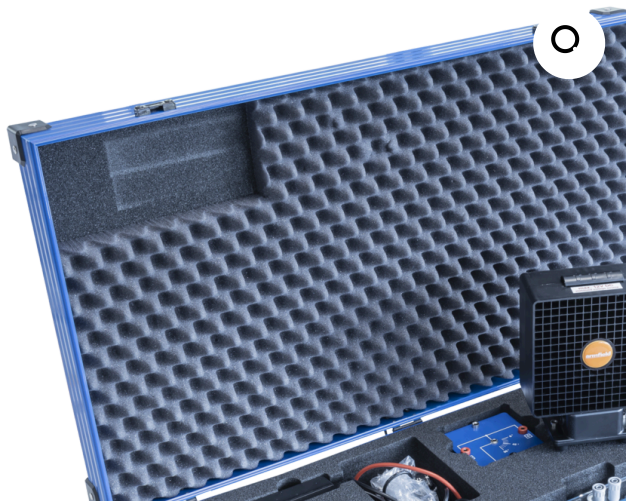




RE12 – Advanced Wind Energy

The RE12 Advanced Wind Energy system provides students with a self-contained modular system, covering the physical fundamentals of wind energy productions, covering wind power fundamentals, controlling a typical wind tunnel and different rotor types.

Using the system students study wind turbine characteristics, turbine performance under load and turbine performance at different wind conditions.

PRODUCT CODE: RE12

Categories: Educational, RE Series - Renewable Energy, RE12 Wind Energy

Tags: Blade Number , Blade Shape , Ohm`S Law , Optimised Working Conditions , RE Series , RE12 , Renewable Energy , Savonius Rotors , Turbine Performance , Wind Conditions

DESCRIPTION

The RE12 Advanced Wind Energy system provides students with a self-contained modular system, covering the physical fundamentals of wind energy productions, covering wind power fundamentals, controlling a typical wind tunnel and different rotor types.

Using the system students study wind turbine characteristics, turbine performance under load and turbine performance at different wind conditions.

This is achieved by practice experimentation designed to analyse the power conversion processes at a wind turbine, characterising and rating of a typical wind turbine, investigating the influence of wind conditions and loads on turbine performance and determining optimized working conditions.

The system is supplied with multiple rotors allowing student to investigate Savonius rotors, blade shape, blade number and operating conditions. Detailed experiments are provided allowing the student to understand the advantages and disadvantages of different rotor types ultimately proving the argument that a three-blade horizontal axis wind turbine is the most efficient design.

TECHNICAL SPECIFICATIONS

- 1 x Wind turbine module Pro
- 1 x Potentiometer module Pro
- 1 x Savonius rotor module Pro
- 1 x Wind rotor set
- 1 x Base unit Professional
- 1 x Wind generator
- 1 x Anemometer Pro
- 2 x Resistor module (triple) Pro
- 3 x Resistor plug element 100 Ohm
- 2 x Resistor plug element 10 Ohm
- 1 x Resistor plug element 33 Ohm
- 1 x AV module
- 1 x Power module
- 1 x Propeller
- 1 x Safety test lead, 50cm, red
- 1 x Safety test lead, 50cm, black
- 1 x Safety test lead, 25cm, red
- 1 x Safety test lead, 25cm, black
- 3 x Safety short-circuit plug, with mid socket
- 1 x Rotational-speed sensor
- 1 x Aluminium case "Wind-Professional"

FEATURES & BENEFITS

- Supplied with 2, 3 and 4 blade rotor of variable pitch and blade design
- Allows configuration of 24 different wind turbines
- Includes Savonius rotor
- Laboratory scale
- Modular design
- Supplied in a self contained aluminium case
- Includes in-depth manual and predefined experiments

Experimental Content

Basic electronic experiments

- Setup of a simple circuit
- Ohm's law
- Series connection of ohmic resistances
- Parallel connection of ohmic resistances
- Start-up and idling behaviour of a motor

Basic wind experiments

- Examine the wind speed behind the rotor
- Energy balance sheet and efficiency of a wind turbine
- Rotational speed and speed ratio of a wind turbine
- Change the turbine voltage by connecting a consumer



Influence of a consumer

- Characteristic curves and rotational speed of a wind turbine
- Influence of the wind speed
- Voltage of a wind turbine dependent on the wind speed
- Rotational speed and output dependent on the wind speed
- Voltage dependent on the wind direction
- Influence of the wind direction
- Rotational speed and output dependent on the wind direction
- Influence of the generator model
- u Voltage dependent on the rotor model
- Rotational speed and output dependent on the rotor model
- Influence of the rotor blade shape
- Voltage dependent on the rotor blade shape
- Rotational speed and output dependent on the rotor blade shape
- Influence of the number of rotor blades
- Voltage dependent on the number of blades
- Rotational speed and output dependent on the number of blades
- Voltage dependent on the rotor blade pitch
- Influence of the rotor blade pitch dependent on the rotor blade pitch
- Start-up speed of a wind turbine dependent on the rotor blade pitch
- Rotational speed and output dependent on the rotor blade pitch

Related curriculums

- Electrical Engineering
- Renewable Energies

Other products in the series

- **RE10:** Advanced Photovoltaic Energy
- **RE14:** Advanced Fuel Cell Technology
- **RE16:** Advanced Thermal Energy
- **RE18:** Advanced Smart Grid Technology
- **RE24:** Advanced Battery Technology

Operational conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface



Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 6Kg

Dimensions

TRAY

- **Length:** 0.640m
- **Width:** 0.165m
- **Height:** 0.370m

Order Codes

RE12: Advanced Wind Energy

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**RE18 –
ADVANCED
SMART GRID
TECHNOLOGY**

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**RE16 –
ADVANCED
THERMAL
ENERGY**

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RE20 – BIOFUEL

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**RE22 –
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**RE24 –
ADVANCED
BATTERY
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